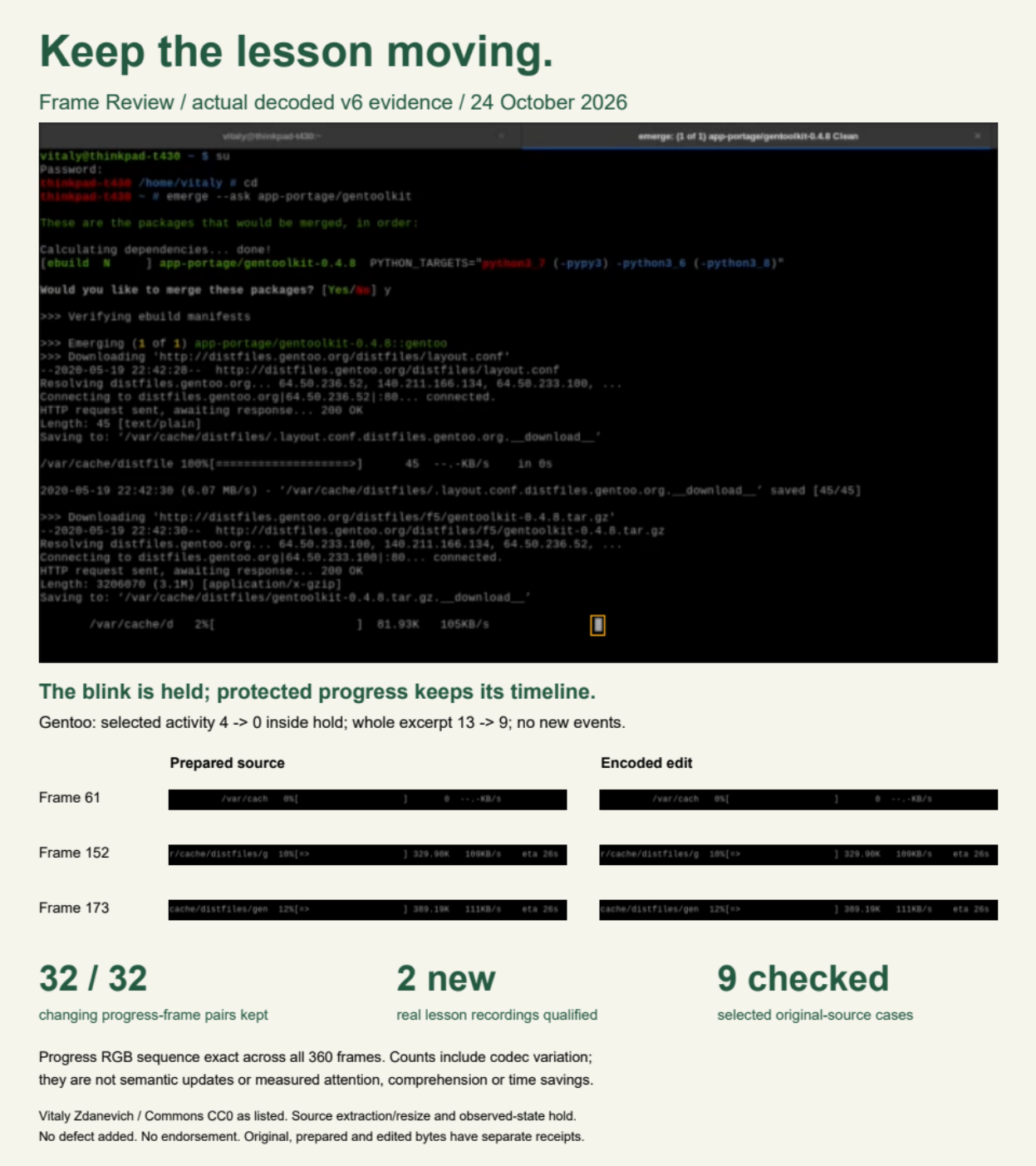
Frame Review: Keep the Lesson

Keep the lesson moving. Calm the idle blink.

A teacher has a finished screen recording and no editable project. One idle caret can keep blinking while the explanation continues. Frame Review lets an editor inspect both observed states, protect the teaching content, and generate a narrowly bounded comparison. The encoded result must preserve the teaching timeline and pass every cut-point check.



Proof of what keeps changing

v6 adds a frame-ordered teaching proof, not a new detector threshold. Each declared region is compared on all 360 source and encoded-edit RGB frames. SHA-256 includes a big-endian frame index and exact RGB bytes in order. Equal activity counts alone cannot pass a reordered or frozen sequence.

In Gentoo, the protected live-progress crop changes on 32 of 112 adjacent frame pairs inside the held interval in both source and edit. All 360 ordered teaching crops match exactly. These changes include compression variation; no semantic update count, readability or comprehension is inferred. Commands above the readout retain 15 changing pairs; the lower margin retains one.

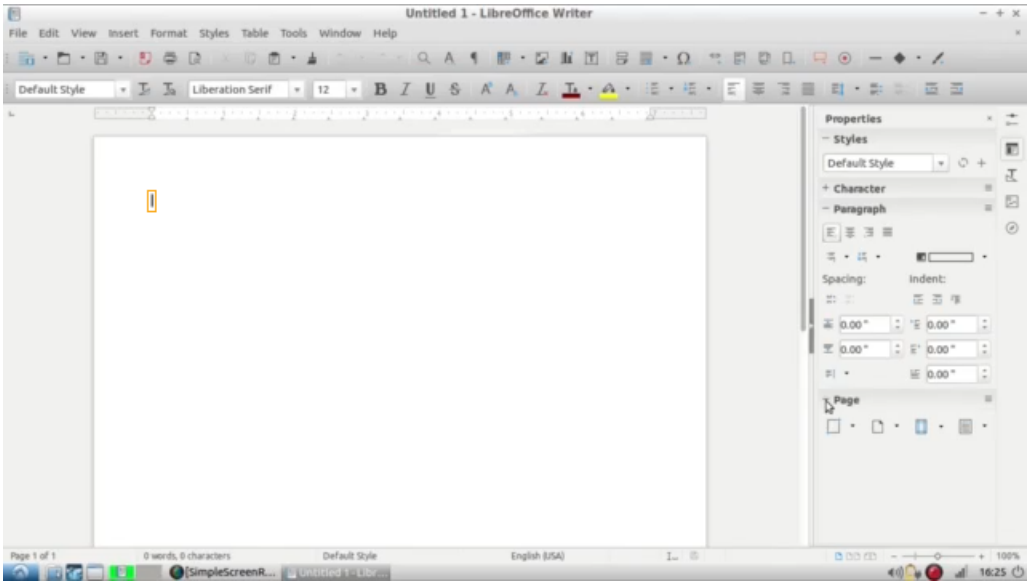
At hold entry, the largest measured teaching change inside the hold, and hold end, the app serves static source/edit PNG pairs. It binds actual PNG hashes, decoded crop RGB hashes, dimensions, input, prediction, approved recipe and output. Proof corruption withholds the image. Selected moments aid inspection; they are not a full-motion trial.

Check	Observed result
Full Linux gate	14 stages pass; 137 backend tests; 91 studio tests, one platform skip
Original HTTP qualifications	Seven previous sources plus two additional recordings; all nine pass independent decode, source recovery, boundary and audio/absence checks
Teaching integrity	Actual HTTP PNG downloads, independent decoded-crop checks and frame-order hashes pass in all nine; mutated PNG and receipt are refused, then restored
Local browser	Untouched Gentoo original upload, explicit plan/approval, seven artifacts, twelve boundary PNGs and eighteen teaching PNGs downloaded and hash checked; reload and actual 375/890 widths pass
Runtime freeze	09c8bd44526e7d1e65ad40ea2528154b3f6489c7659510bb61b5fcd5fddfa123

No detector/cut threshold, 120-second worker CPU budget, original-size cap, profile or preservation tolerance was loosened. The new teaching proof uses the existing decoded-frame pass. Unit tests separately exercise teaching freezes, reordered frames, missing frames, bounded moments and actual PNG pixels.

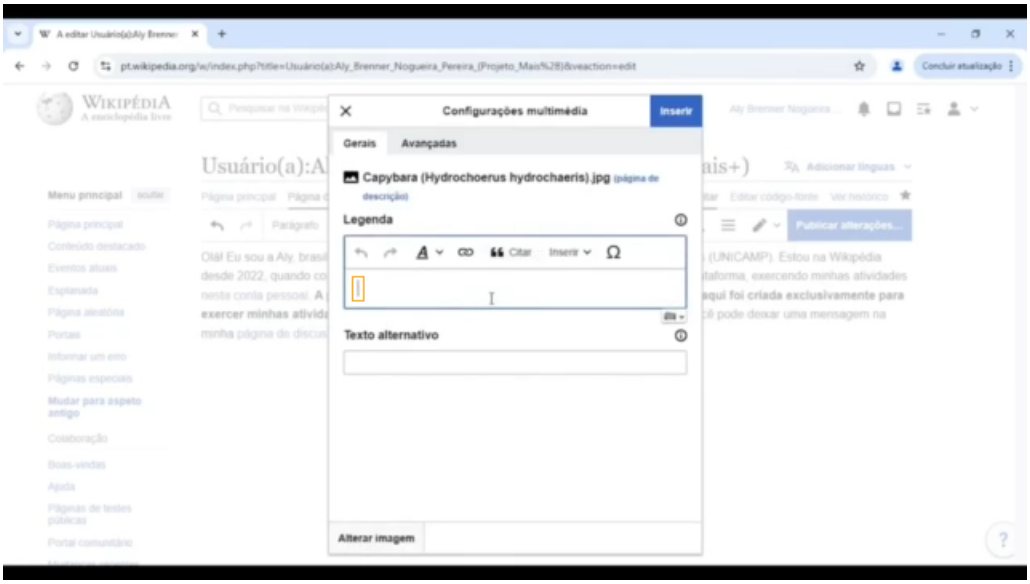
Two additional real-source repairs

Writer: a blank-page explanation



Original start 150 s; prepared 12 s / 360 frames. Held frames [1, 119]; observed source reference 0. Inside hold 6 -> 0; full excerpt 15 -> 9; zero newly timed events. All teaching/native/audio/timing and encoded boundary gates pass.

Wikipedia: an image-caption explanation



Original start 236 s; prepared 12 s / 360 frames. Held frames [244, 354]; observed source reference 346. Inside hold 5 -> 0; full excerpt 17 -> 11; zero newly timed events. All teaching/native/audio/timing and encoded boundary gates pass.

Writer: Free I.T. Athens, CC BY 3.0 as listed. Wikipedia: exact creator and CC BY-SA 4.0 metadata in sources.json and examples/CREDITS.json; share-alike conditions apply to the adapted output. Images above are actual reviewed source states with selection overlays. No visual defect was authored. A creator may still prefer the original caret.

Every outcome stays in the evidence

Eleven additional originals / thirty-three fixed windows were screened. The first seven sources / 21 windows used frozen v5. Four more sources / 12 windows were selected after that screen and used v6, whose detector and boundary policy are identical. Both are convenience exploration, not held-out recall or a success rate. Three Toybox recordings share an author.

New source group	Retained outcome
Toybox prebuilt and compiling	Six no-proposal windows
Toybox implementation	Two no-proposal; one stationary proposal touches command-menu teaching content; protected plan refuses
Writer basics	One newly qualified reading-pause caret; one rapid-only window declined; one title/letter overlap refuses
Calc import	Two no-proposal; one caret/dataset overlap refuses with data protected
Charts and Pandoc	Six no-proposal windows
Gerrit setup	Three no-proposal windows
Equation annotation	Two rapid-only windows declined; one no-proposal
Wikipedia user page	Two rapid-only windows declined; one mixed window with the separately reviewed stationary caption-caret repair
Wikipedia sandbox	Three rapid-only windows declined

All three actual teaching-overlap plans refuse before approval, exact original and prepared bytes are recovered, and edited downloads are withheld. A first Calc protection request hit the 40-character label limit; the harness was corrected and only that failed case retried. The initial failure remains recorded. This was a test configuration error, not a repaired runtime defect.

The prior six-source, same-72-second comparison remains unchanged: source 90, untuned 5x5/sigma2 local blur 91, prior longer hold 54, matched hold 56. Matched holds introduce no new threshold events; prior holds do in five cases. The new holds total 17.6 seconds rather than 24. Full traces and negative results remain in v5 evidence. A lower count alone does not establish a good edit.

New cuts are checked after actual encoding: source/edit selected and influence RGB MAE, RMS, max at entry, return and loop, plus frozen selected-mean diagnostics. First/last clip frames stay unchanged; no new eight-code threshold event is allowed. Existing strong source transitions may remain. The actual former bad Jupyter hold is encoded and refused in the full gate.

Reproduce and assess the impact hypothesis

Build on Linux: `docker build -t frame-review .` Run in a fresh container: `docker run --name frame-review-check --network none frame-review python studio/run.py --verify /tmp/qualification`. Copy receipts with docker cp. Serve: `docker run --rm --network host frame-review python studio/run.py --serve --port 8877`. Windows source/UI checks alone do not qualify media.

Exact source URLs, SHA-256, listed authors/licenses and adaptations are in evidence/natural-v6/sources.json. The two new case configurations and actual HTTP/independent-decode harnesses are in docs/natural-v6/. Individual nine-source receipts include cuts, teaching-region ordered digests, measured variation, source recovery, artifact tamper refusal and actual output hashes. Two local HTTP comparison downloads and credits are bundled; cloud repeats are labeled separately.

AWS: existing ECS/Fargate default-lesson-rescue:11. Image sha256:957fdbd114072441fdc1c948fa0dc02eb5011b0d0bf7757ce83908c8308ae45f. Three original-upload browser jobs, 36 boundary PNGs, 60 teaching PNGs, ownership 404s, reloads and actual 12-second downloaded playbacks pass. Actual 375/890 widths were checked on caption; other two were observed at 1280. These repeat local cases. Playback is muted; no listening claim.

Gentoo and Writer prepared/edit bytes match local qualification. Caption preparation video packets/native samples differ across hosts; audio packets/PCM match. Its actual cloud pair independently passes native mapping, outside-region/interval, all 360 teaching RGB frames, actual proof PNG pixels, audio/timing and boundary/event checks. Cause of cross-host variation is unestablished; no universal preparation-byte identity claim. A first standalone diagnostic omitted the frozen explicit RGB conversion and failed PNG comparison; the failed result and corrected-contract rerun are retained. See cloud-caption-independent.json and cloud-diagnostic-negative.json.

The live service uses one 1-vCPU/2-GiB task, one worker and up to eight retained jobs. No application task role is added. Originals are limited to 64 MiB and supported SDR tracks. Preparation extracts 1-12 seconds, resizes/letterboxes to 640x360/30 fps and converts optional audio once. Original and prepared bytes are separate; legibility can be reduced. Exact edit checks reference the prepared source. Jobs expire after 24 hours or can disappear on restart.

The impact hypothesis is a teacher/editor rescuing a small explanation without re-recording the lesson. The current evidence establishes bounded local change, exact declared teaching samples, source recovery and reproducibility. It does not establish creator preference, time saved, attention, learning, accessibility, safety or adoption. Next evidence should be consenting creator tasks with blinded original/edit preference and timed rescue attempts; those trials have not run.

No full-lesson splice-back, semantic motion tracking, original-resolution preservation guarantee or automatic publication decision is implemented. Source metadata is attribution, not independent rights clearance or endorsement. The frozen earlier engine, negative evidence and controlled added-corner examples remain versioned separately. The existing 2:46 neural-narrated demo shows an earlier workflow; Joseph’s real voice is deferred. No special-award capability, judge score or win is claimed.